Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Machado Dairy Farms #3

Physical address of dairy:

Number and Street	City	County	Zip Code
11685 E Mariposa RD	Stockton	San Joaquin	95215

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0181-0070-0001-0000

B. OPERATORS

Machado, Donald

Operator name: Machado, Donald

Telephone no.: Landline: ; Cellular: (209) 602-3473

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 4430	Manteca	CA	95337

This operator is responsible for paying permit fees.

Machado Dairy Farms #3

Operator name: Machado Dairy Farms #3

Telephone no.: Landline: ; Cellular: (209) 602-2473

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 4430	Manteca	CA	95337

C. OWNERS

Machado, Donald

Legal owner name:	Machado, Donald	Telephone no.:		(209) 602-3473
			Landline	Cellular
P.O. Box 4430		Manteca	CA	95337
Mailing Address Number and Street		City	State	Zip Code

This owner is responsible for paying permit fees.

Machado Dairy Farms #3

Legal owner name:	Machado Dairy Farms #3	Telephone no.:		(209) 602-2473
			Landline	Cellular
P.O. Box 4430		Manteca	CA	95337
Mailing Address Number and Street		City	State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	250	50	0	0	0
Number under roof	1,100	0	0	0	0	0
Maximum number	1,100	250	50	0	0	0
Average number	1,100	250	50	0	0	0
Avg live weight (lbs)	1,400	1,450	900	0		

Predominant milk cow breed: Holstein

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 32,375.54 *tons per reporting period*

Total nitrogen from manure: 419,288.11 *lbs per reporting period*

After ammonia losses (30% loss applied): 293,501.68 *lbs per reporting period*

Total phosphorus from manure: 69,281.12 *lbs per reporting period*

Total potassium from manure: 210,660.15 *lbs per reporting period*

Total salt from manure: 575,422.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,312,000 *gallons*

Total nitrogen generated: 25,648.15 *lbs*

Total phosphorus generated: 2,151.33 *lbs*

Total potassium generated: 44,858.71 *lbs*

Total salt generated: 234,632.03 *lbs*

	3,312,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	3,312,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
3Dp1	Surface water
3Dp2	Ground water
3Dp3	Ground water
3Dp4	Ground water
3Dp5	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/08/2023	Compost	3,760.00 *ton*	Dry-weight	12.2		23,700.00	9,040.00	48,200.00		46.50

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	156,480.67	59,687.14	318,243.39	3,070,190.40
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	156,480.67	59,687.14	318,243.39	3,070,190.40

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
6-1	17	17	2	both	X181-X030-XX02-XXXX
6-11	38	38	2	both	X181-X070-XX01-XXXX
6-12	10	10	2	both	X181-X070-XX01-XXXX
6-2	18	18	2	both	X181-X070-XX01-XXXX
6-3	40	40	2	both	X181-X030-XX02-XXXX
6-4	5	5	2	manure	X181-X030-XX02-XXXX
6-5	32	32	2	both	X181-X030-XX02-XXXX
6-6	39	39	2	both	X181-X030-XX02-XXXX
6-7	49	49	2	both	X181-X030-XX02-XXXX
6-8	15	15	2	both	X181-X070-XX01-XXXX
6-9	18	18	2	both	X181-X070-XX01-XXXX
Totals for areas that were used for application	281	281	22		
Totals for areas that were not used for application					
Land application area totals	281	281	22		

B. CROPS AND HARVESTS

6-1

Field name: 6-1

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	250.00 *ton*	Dry-weight		63.4	17,300.00	3,680.00	31,600.00		13.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.71	186.23	39.61	340.16	1,431.71

6-1

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	440.00 *ton*	Dry-weight		64.8	17,200.00	3,450.00	17,800.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	25.88	313.40	62.86	324.34	1,548.80

6-11

Field name: 6-11

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	570.00 *ton*	Dry-weight		67.3	19,700.00	3,540.00	26,500.00		15.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	193.26	34.73	259.97	1,540.17

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	1,000.00 *ton*	Dry-weight		70.9	18,200.00	3,270.00	20,700.00		9.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.32	278.75	50.08	317.04	1,485.63

6-12

Field name: 6-12

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	150.00 *ton*	Dry-weight		65.5	19,700.00	3,520.00	27,000.00		16.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	203.90	36.43	279.45	1,656.00

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	260.00 *ton*	Dry-weight		68.6	17,700.00	3,380.00	24,400.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.00	289.01	55.19	398.40	1,665.46

6-2

Field name: 6-2

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	260.00 *ton*	Dry-weight		65.7	16,900.00	4,280.00	33,200.00		12.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.44	167.46	42.41	328.98	1,278.25

6-2

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	470.00 *ton*	Dry-weight		67.7	18,300.00	3,320.00	19,200.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.11	308.68	56.00	323.86	1,400.03

6-3

Field name: 6-3

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	500.00 *ton*	Dry-weight		68.8	24,000.00	4,420.00	29,600.00		15.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.50	187.20	34.48	230.88	1,177.80

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	1,050.00 *ton*	Dry-weight		59.8	18,100.00	3,780.00	18,900.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.25	382.00	79.78	398.88	1,878.35

6-4

Field name: 6-4

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 5 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	70.00 *ton*	Dry-weight		71.5	21,800.00	4,290.00	36,400.00		16.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.00	173.96	34.23	290.47	1,300.74

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 5 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	130.00 *ton*	Dry-weight		71.5	20,100.00	4,690.00	19,300.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.00	297.88	69.51	286.03	1,363.44

6-5

Field name: 6-5

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	470.00 *ton*	Dry-weight		65.8	23,600.00	5,060.00	39,200.00		15.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.69	237.09	50.83	393.81	1,506.94

6-5

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	830.00 *ton*	Dry-weight		75.2	17,800.00	2,730.00	22,100.00		9.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	25.94	229.00	35.12	284.32	1,196.45

6-6

Field name: 6-6

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 39 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	590.00 *ton*	Dry-weight		66.0	25,400.00	4,950.00	40,400.00		15.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.13	261.29	50.92	415.60	1,553.36

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	1,020.00 *ton*	Dry-weight		68.4	17,900.00	2,960.00	18,200.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.15	295.87	48.93	300.83	1,487.63

6-7

Field name: 6-7

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 49 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/23/2023	740.00 *ton*	Dry-weight		70.9	26,900.00	5,020.00	30,500.00		15.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.10	236.43	44.12	268.08	1,344.78

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 49 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	1,280.00 *ton*	Dry-weight		69.5	17,600.00	3,810.00	17,900.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.12	280.45	60.71	285.23	1,497.86

6-8

Field name: 6-8

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	230.00 *ton*	Dry-weight		65.9	22,800.00	4,270.00	35,700.00		14.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.33	238.43	44.65	373.33	1,526.77

6-8

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	400.00 *ton*	Dry-weight		63.1	17,900.00	3,340.00	15,600.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.67	352.27	65.73	307.01	1,692.48

6-9

Field name: 6-9

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 18 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	270.00 *ton*	Dry-weight		70.5	25,800.00	3,930.00	30,100.00		15.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	15.00	228.33	34.78	266.39	1,354.05

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 18 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	470.00 *ton*	Dry-weight		62.4	16,500.00	3,520.00	17,800.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	208.00	39.00	171.60	0.00
Total actual harvest content	26.11	323.99	69.12	349.51	1,708.29

NUTRIENT BUDGET

A. LAND APPLICATIONS

6-1 - 10/29/2022: Oats, silage-soft dough

Field name: 6-1

Crop: Oats, silage-soft dough — Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	200.53	91.83	248.63	0.00	60.00 *ton*
Application event totals		200.53	91.83	248.63	0.00	

6-1 - 05/28/2023: Corn, silage

Field name: 6-1

Crop: Corn, silage — Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	56.14	30.91	42.96	0.00	30.00 *ton*
Application event totals		56.14	30.91	42.96	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	26.83	0.00	0.00	710.40	2,160,000.00 *gal*
Application event totals		26.83	0.00	0.00	710.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	26.83	0.00	0.00	710.40	2,160,000.00 *gal*
Application event totals		26.83	0.00	0.00	710.40	

6-1 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	24.59	0.00	0.00	651.20	1,980,000.00 *gal*
Application event totals		24.59	0.00	0.00	651.20	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	43.68	3.36	77.28	394.91	96,000.00 *gal*
Application event totals		43.68	3.36	77.28	394.91	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	24.59	0.00	0.00	651.20	1,980,000.00 *gal*
Application event totals		24.59	0.00	0.00	651.20	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	24.59	0.00	0.00	651.20	1,980,000.00 *gal*
Application event totals		24.59	0.00	0.00	651.20	

09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	24.59	0.00	0.00	651.20	1,980,000.00 *gal*
Application event totals		24.59	0.00	0.00	651.20	

09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	24.59	0.00	0.00	651.20	1,980,000.00 *gal*
Application event totals		24.59	0.00	0.00	651.20	

6-11 - 10/29/2022: Oats, silage-soft dough

6-11 - 10/29/2022: Oats, silage-soft dough

Field name: 6-11

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	224.28	102.71	278.08	0.00	150.00 *ton*
Application event totals		224.28	102.71	278.08	0.00	

6-11 - 05/28/2023: Corn, silage

Field name: 6-11

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	83.72	46.10	64.06	0.00	100.00 *ton*
Application event totals		83.72	46.10	64.06	0.00	

05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	6.25	0.00	0.00	231.24	4,680,000.00 *gal*
Application event totals		6.25	0.00	0.00	231.24	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	6.25	0.00	0.00	231.24	4,680,000.00 *gal*
Application event totals		6.25	0.00	0.00	231.24	

6-11 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	5.77	0.00	0.00	213.46	4,320,000.00 *gal*
Application event totals		5.77	0.00	0.00	213.46	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	88.44	10.06	147.05	850.03	432,000.00 *gal*
Application event totals		88.44	10.06	147.05	850.03	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	5.77	0.00	0.00	213.46	4,320,000.00 *gal*
Application event totals		5.77	0.00	0.00	213.46	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	5.77	0.00	0.00	213.46	4,320,000.00 *gal*
Application event totals		5.77	0.00	0.00	213.46	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	87.94	6.76	155.59	795.01	432,000.00 *gal*
Application event totals		87.94	6.76	155.59	795.01	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	5.77	0.00	0.00	213.46	4,320,000.00 *gal*
Application event totals		5.77	0.00	0.00	213.46	

6-11 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	5.77	0.00	0.00	213.46	4,320,000.00 *gal*
Application event totals		5.77	0.00	0.00	213.46	

6-12 - 10/29/2022: Oats, silage-soft dough

Field name: 6-12

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	227.27	104.08	281.78	0.00	40.00 *ton*
Application event totals		227.27	104.08	281.78	0.00	

6-12 - 05/28/2023: Corn, silage

Field name: 6-12

Crop: Corn, silage

Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	127.25	70.07	97.37	0.00	40.00 *ton*
Application event totals		127.25	70.07	97.37	0.00	

6-12 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	7.31	0.00	0.00	270.38	1,440,000.00 *gal*
Application event totals		7.31	0.00	0.00	270.38	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	6.39	0.00	0.00	236.58	1,260,000.00 *gal*
Application event totals		6.39	0.00	0.00	236.58	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	6.39	0.00	0.00	236.58	1,260,000.00 *gal*
Application event totals		6.39	0.00	0.00	236.58	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	149.36	16.98	248.35	1,435.61	192,000.00 *gal*
Application event totals		149.36	16.98	248.35	1,435.61	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	6.39	0.00	0.00	236.58	1,260,000.00 *gal*
Application event totals		6.39	0.00	0.00	236.58	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	6.39	0.00	0.00	236.58	1,260,000.00 *gal*
Application event totals		6.39	0.00	0.00	236.58	

6-12 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	6.39	0.00	0.00	236.58	1,260,000.00 *gal*
Application event totals		6.39	0.00	0.00	236.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp1	Surface water	6.39	0.00	0.00	236.58	1,260,000.00 *gal*
Application event totals		6.39	0.00	0.00	236.58	

6-2 - 10/29/2022: Oats, silage-soft dough

Field name: 6-2

Crop: Oats, silage-soft dough Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	220.96	101.19	273.96	0.00	70.00 *ton*
Application event totals		220.96	101.19	273.96	0.00	

6-2 - 05/28/2023: Corn, silage

Field name: 6-2

Crop: Corn, silage Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

6-2 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	70.70	38.93	54.10	0.00	40.00 *ton*
Application event totals		70.70	38.93	54.10	0.00	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	25.34	0.00	0.00	670.94	2,160,000.00 *gal*
Application event totals		25.34	0.00	0.00	670.94	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	25.34	0.00	0.00	670.94	2,160,000.00 *gal*
Application event totals		25.34	0.00	0.00	670.94	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	23.22	0.00	0.00	615.03	1,980,000.00 *gal*
Application event totals		23.22	0.00	0.00	615.03	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	41.26	3.17	72.99	372.97	96,000.00 *gal*
Application event totals		41.26	3.17	72.99	372.97	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	23.22	0.00	0.00	615.03	1,980,000.00 *gal*
Application event totals		23.22	0.00	0.00	615.03	

6-2 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	23.22	0.00	0.00	615.03	1,980,000.00 *gal*
Application event totals		23.22	0.00	0.00	615.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	23.22	0.00	0.00	615.03	1,980,000.00 *gal*
Application event totals		23.22	0.00	0.00	615.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp5	Ground water	23.22	0.00	0.00	615.03	1,980,000.00 *gal*
Application event totals		23.22	0.00	0.00	615.03	

6-3 - 10/29/2022: Oats, silage-soft dough

Field name: 6-3

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	213.06	97.57	264.17	0.00	150.00 *ton*
Application event totals		213.06	97.57	264.17	0.00	

6-3 - 05/28/2023: Corn, silage

Field name: 6-3

Crop: Corn, silage

Plant date: 05/28/2023

6-3 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	79.53	43.79	60.86	0.00	100.00 *ton*
Application event totals		79.53	43.79	60.86	0.00	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	27.44	0.00	0.00	604.60	5,040,000.00 *gal*
Application event totals		27.44	0.00	0.00	604.60	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	25.48	0.00	0.00	561.41	4,680,000.00 *gal*
Application event totals		25.48	0.00	0.00	561.41	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	25.48	0.00	0.00	561.41	4,680,000.00 *gal*
Application event totals		25.48	0.00	0.00	561.41	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	25.48	0.00	0.00	561.41	4,680,000.00 *gal*
Application event totals		25.48	0.00	0.00	561.41	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	37.13	2.86	65.69	335.67	192,000.00 *gal*
Application event totals		37.13	2.86	65.69	335.67	

6-3 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	25.48	0.00	0.00	561.41	4,680,000.00 *gal*
Application event totals		25.48	0.00	0.00	561.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	25.48	0.00	0.00	561.41	4,680,000.00 *gal*
Application event totals		25.48	0.00	0.00	561.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	25.48	0.00	0.00	561.41	4,680,000.00 *gal*
Application event totals		25.48	0.00	0.00	561.41	

6-4 - 10/29/2022: Oats, silage-soft dough

Field name: 6-4

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	227.27	104.08	281.78	0.00	20.00 *ton*
Application event totals		227.27	104.08	281.78	0.00	

6-4 - 05/28/2023: Corn, silage

Field name: 6-4

Crop: Corn, silage

Plant date: 05/28/2023

6-4 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	127.25	70.07	97.37	0.00	20.00 *ton*
Application event totals		127.25	70.07	97.37	0.00	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	31.36	0.00	0.00	690.97	720,000.00 *gal*
Application event totals		31.36	0.00	0.00	690.97	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	31.36	0.00	0.00	690.97	720,000.00 *gal*
Application event totals		31.36	0.00	0.00	690.97	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	23.52	0.00	0.00	518.22	540,000.00 *gal*
Application event totals		23.52	0.00	0.00	518.22	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	23.52	0.00	0.00	518.22	540,000.00 *gal*
Application event totals		23.52	0.00	0.00	518.22	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	23.52	0.00	0.00	518.22	540,000.00 *gal*
Application event totals		23.52	0.00	0.00	518.22	

6-4 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	23.52	0.00	0.00	518.22	540,000.00 *gal*
Application event totals		23.52	0.00	0.00	518.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp3	Ground water	23.52	0.00	0.00	518.22	540,000.00 *gal*
Application event totals		23.52	0.00	0.00	518.22	

6-5 - 10/29/2022: Oats, silage-soft dough

Field name: 6-5

Crop: Oats, silage-soft dough Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	248.57	113.83	308.20	0.00	140.00 *ton*
Application event totals		248.57	113.83	308.20	0.00	

6-5 - 05/28/2023: Corn, silage

Field name: 6-5

Crop: Corn, silage Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

6-5 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	139.18	76.64	106.50	0.00	140.00 *ton*
Application event totals		139.18	76.64	106.50	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	13.52	0.00	0.00	473.16	4,320,000.00 *gal*
Application event totals		13.52	0.00	0.00	473.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	13.52	0.00	0.00	473.16	4,320,000.00 *gal*
Application event totals		13.52	0.00	0.00	473.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.27	0.00	0.00	394.30	3,600,000.00 *gal*
Application event totals		11.27	0.00	0.00	394.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.27	0.00	0.00	394.30	3,600,000.00 *gal*
Application event totals		11.27	0.00	0.00	394.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	81.23	6.25	143.70	734.28	336,000.00 *gal*
Application event totals		81.23	6.25	143.70	734.28	

6-5 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.27	0.00	0.00	394.30	3,600,000.00 *gal*
Application event totals		11.27	0.00	0.00	394.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.27	0.00	0.00	394.30	3,600,000.00 *gal*
Application event totals		11.27	0.00	0.00	394.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.27	0.00	0.00	394.30	3,600,000.00 *gal*
Application event totals		11.27	0.00	0.00	394.30	

6-6 - 10/29/2022: Oats, silage-soft dough

Field name: 6-6

Crop: Oats, silage-soft dough Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	218.53	100.07	270.95	0.00	150.00 *ton*
Application event totals		218.53	100.07	270.95	0.00	

6-6 - 05/28/2023: Corn, silage

Field name: 6-6

Crop: Corn, silage Plant date: 05/28/2023

6-6 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	146.83	80.85	112.35	0.00	180.00 *ton*
Application event totals		146.83	80.85	112.35	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	12.32	0.00	0.00	431.37	4,800,000.00 *gal*
Application event totals		12.32	0.00	0.00	431.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	10.48	0.00	0.00	366.67	4,080,000.00 *gal*
Application event totals		10.48	0.00	0.00	366.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	10.48	0.00	0.00	366.67	4,080,000.00 *gal*
Application event totals		10.48	0.00	0.00	366.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	76.17	5.86	134.75	688.55	384,000.00 *gal*
Application event totals		76.17	5.86	134.75	688.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	10.48	0.00	0.00	366.67	4,080,000.00 *gal*
Application event totals		10.48	0.00	0.00	366.67	

6-6 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	10.48	0.00	0.00	366.67	4,080,000.00 *gal*
Application event totals		10.48	0.00	0.00	366.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	10.48	0.00	0.00	366.67	4,080,000.00 *gal*
Application event totals		10.48	0.00	0.00	366.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	10.48	0.00	0.00	366.67	4,080,000.00 *gal*
Application event totals		10.48	0.00	0.00	366.67	

6-7 - 10/29/2022: Oats, silage-soft dough

Field name: 6-7

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	231.91	106.20	287.53	0.00	200.00 *ton*
Application event totals		231.91	106.20	287.53	0.00	

6-7 - 05/28/2023: Corn, silage

Field name: 6-7

Crop: Corn, silage

Plant date: 05/28/2023

6-7 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	110.37	60.77	84.46	0.00	170.00 *ton*
Application event totals		110.37	60.77	84.46	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	14.71	0.00	0.00	515.01	7,200,000.00 *gal*
Application event totals		14.71	0.00	0.00	515.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.77	0.00	0.00	412.00	5,760,000.00 *gal*
Application event totals		11.77	0.00	0.00	412.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.77	0.00	0.00	412.00	5,760,000.00 *gal*
Application event totals		11.77	0.00	0.00	412.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.77	0.00	0.00	412.00	5,760,000.00 *gal*
Application event totals		11.77	0.00	0.00	412.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	90.94	6.99	160.88	822.05	576,000.00 *gal*
Application event totals		90.94	6.99	160.88	822.05	

6-7 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.77	0.00	0.00	412.00	5,760,000.00 *gal*
Application event totals		11.77	0.00	0.00	412.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.77	0.00	0.00	412.00	5,760,000.00 *gal*
Application event totals		11.77	0.00	0.00	412.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.77	0.00	0.00	412.00	5,760,000.00 *gal*
Application event totals		11.77	0.00	0.00	412.00	

6-8 - 10/29/2022: Oats, silage-soft dough

Field name: 6-8

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	227.27	104.08	281.78	0.00	60.00 *ton*
Application event totals		227.27	104.08	281.78	0.00	

6-8 - 05/28/2023: Corn, silage

Field name: 6-8

Crop: Corn, silage

Plant date: 05/28/2023

6-8 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	84.84	46.71	64.92	0.00	40.00 *ton*
Application event totals		84.84	46.71	64.92	0.00	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	12.82	0.00	0.00	448.63	1,920,000.00 *gal*
Application event totals		12.82	0.00	0.00	448.63	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	12.82	0.00	0.00	448.63	1,920,000.00 *gal*
Application event totals		12.82	0.00	0.00	448.63	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.22	0.00	0.00	392.55	1,680,000.00 *gal*
Application event totals		11.22	0.00	0.00	392.55	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.22	0.00	0.00	392.55	1,680,000.00 *gal*
Application event totals		11.22	0.00	0.00	392.55	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	148.53	11.42	262.77	1,342.68	288,000.00 *gal*
Application event totals		148.53	11.42	262.77	1,342.68	

6-8 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.22	0.00	0.00	392.55	1,680,000.00 *gal*
Application event totals		11.22	0.00	0.00	392.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.22	0.00	0.00	392.55	1,680,000.00 *gal*
Application event totals		11.22	0.00	0.00	392.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.22	0.00	0.00	392.55	1,680,000.00 *gal*
Application event totals		11.22	0.00	0.00	392.55	

6-9 - 10/29/2022: Oats, silage-soft dough

Field name: 6-9

Crop: Oats, silage-soft dough Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	220.96	101.19	273.96	0.00	70.00 *ton*
Application event totals		220.96	101.19	273.96	0.00	

6-9 - 05/28/2023: Corn, silage

Field name: 6-9

Crop: Corn, silage Plant date: 05/28/2023

6-9 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Compost	Compost	106.05	58.39	81.14	0.00	60.00 *ton*
Application event totals		106.05	58.39	81.14	0.00	

06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	12.68	0.00	0.00	443.95	2,280,000.00 *gal*
Application event totals		12.68	0.00	0.00	443.95	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	12.68	0.00	0.00	443.95	2,280,000.00 *gal*
Application event totals		12.68	0.00	0.00	443.95	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.35	0.00	0.00	397.22	2,040,000.00 *gal*
Application event totals		11.35	0.00	0.00	397.22	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.35	0.00	0.00	397.22	2,040,000.00 *gal*
Application event totals		11.35	0.00	0.00	397.22	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	123.77	9.52	218.97	1,118.90	288,000.00 *gal*
Application event totals		123.77	9.52	218.97	1,118.90	

6-9 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.35	0.00	0.00	397.22	2,040,000.00 *gal*
Application event totals		11.35	0.00	0.00	397.22	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.35	0.00	0.00	397.22	2,040,000.00 *gal*
Application event totals		11.35	0.00	0.00	397.22	

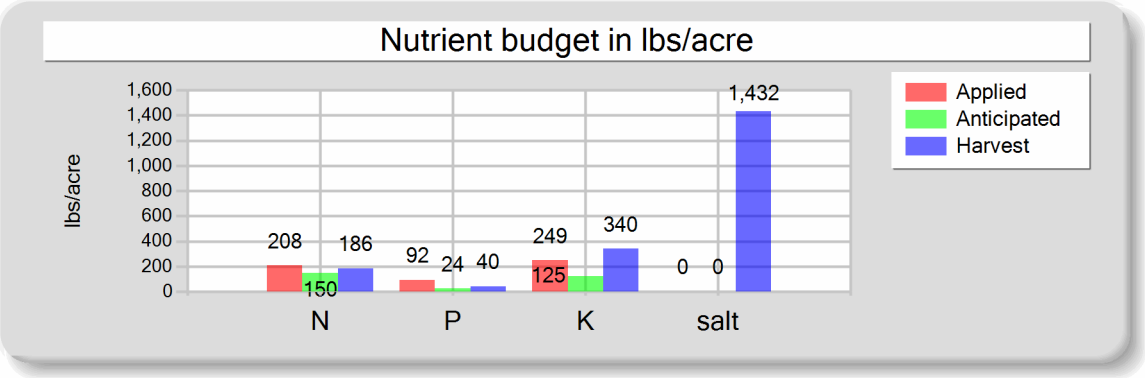
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
3Dp4	Ground water	11.35	0.00	0.00	397.22	2,040,000.00 *gal*
Application event totals		11.35	0.00	0.00	397.22	

B. NUTRIENT BUDGET

6-1 - 10/29/2022: Oats, silage-soft dough

Field name: 6-1 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	200.53	91.83	248.63	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.53	91.83	248.63	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	186.23	39.61	340.16	1,431.71
Nutrient balance	21.30	52.22	-91.53	-1,431.71
Applied to removed ratio	1.11	2.32	0.73	0.00

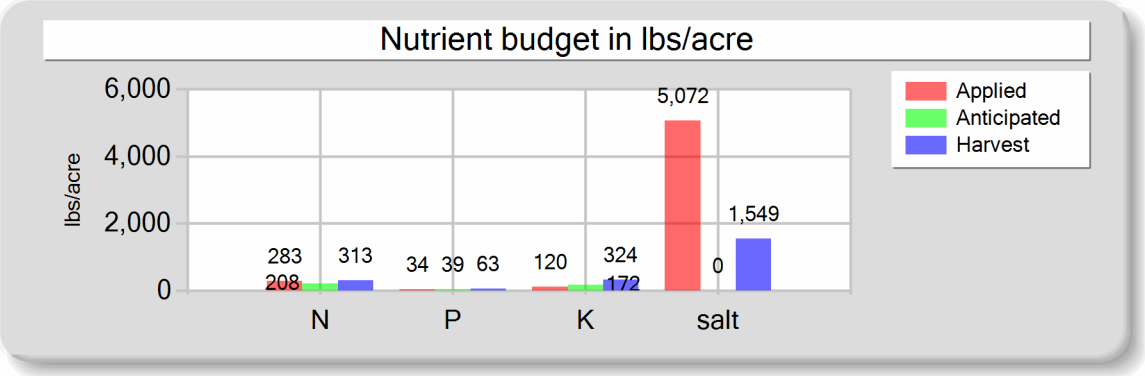
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-1 - 05/28/2023: Corn, silage

Field name: 6-1 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	56.14	30.91	42.96	0.00
Process wastewater	43.68	3.36	77.28	394.91
Fresh water	176.60	0.00	0.00	4,676.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	283.43	34.27	120.24	5,071.74
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	313.40	62.86	324.34	1,548.80
Nutrient balance	-29.98	-28.59	-204.09	3,522.94
Applied to removed ratio	0.90	0.55	0.37	3.27

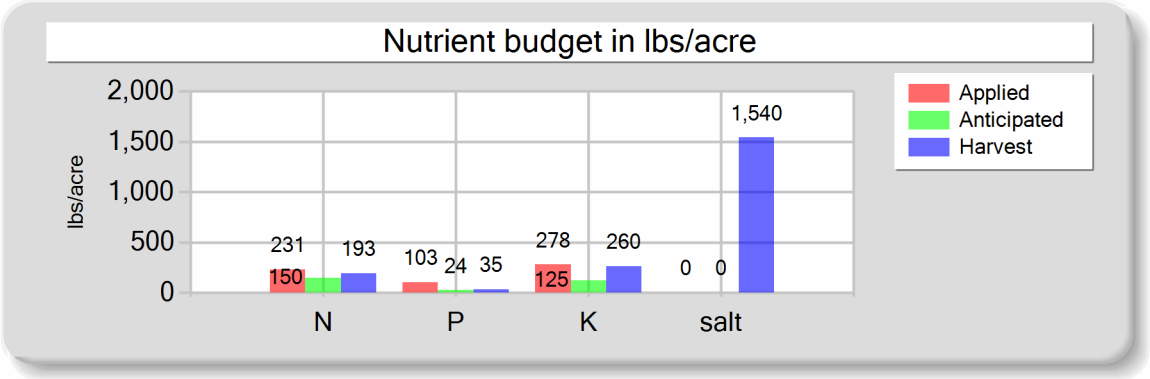
Fresh water applied	
14,220,000.00	*gallons*
523.67	*acre-inches*
30.80	*inches/acre*

Process wastewater applied	
96,000.00	*gallons*
3.54	*acre-inches*
0.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

6-11 - 10/29/2022: Oats, silage-soft dough

Field name: 6-11 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	224.28	102.71	278.08	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.28	102.71	278.08	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	193.26	34.73	259.97	1,540.17
Nutrient balance	38.02	67.98	18.11	-1,540.17
Applied to removed ratio	1.20	2.96	1.07	0.00

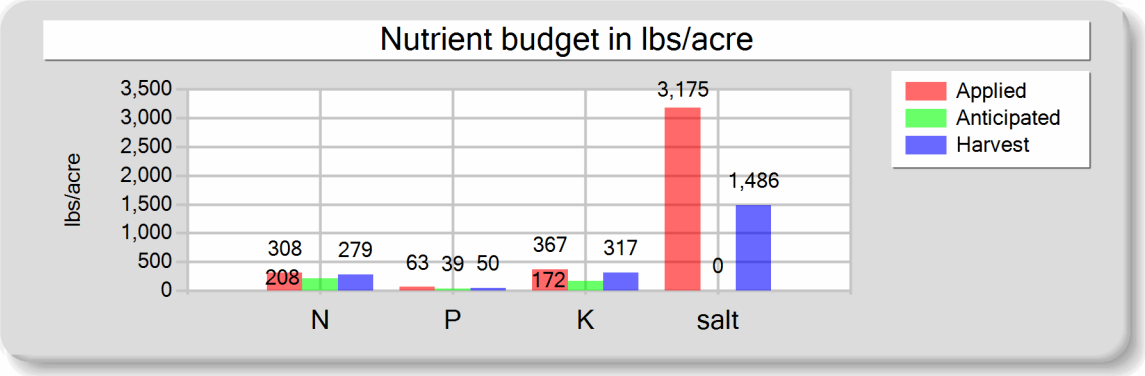
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-11 - 05/28/2023: Corn, silage

Field name: 6-11 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	83.72	46.10	64.06	0.00
Process wastewater	176.38	16.82	302.63	1,645.04
Fresh water	41.34	0.00	0.00	1,529.77
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	308.44	62.92	366.69	3,174.81
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	278.75	50.08	317.04	1,485.63
Nutrient balance	29.69	12.83	49.66	1,689.18
Applied to removed ratio	1.11	1.26	1.16	2.14

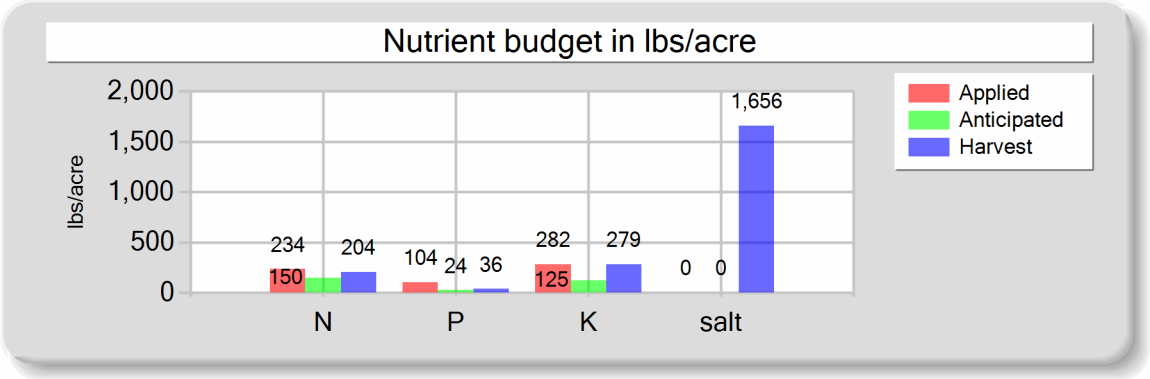
Fresh water applied	
30,960,000.00	*gallons*
1,140.15	*acre-inches*
30.00	*inches/acre*

Process wastewater applied	
864,000.00	*gallons*
31.82	*acre-inches*
0.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

6-12 - 10/29/2022: Oats, silage-soft dough

Field name: 6-12 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	227.27	104.08	281.78	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.27	104.08	281.78	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	203.90	36.43	279.45	1,656.00
Nutrient balance	30.37	67.64	2.33	-1,656.00
Applied to removed ratio	1.15	2.86	1.01	0.00

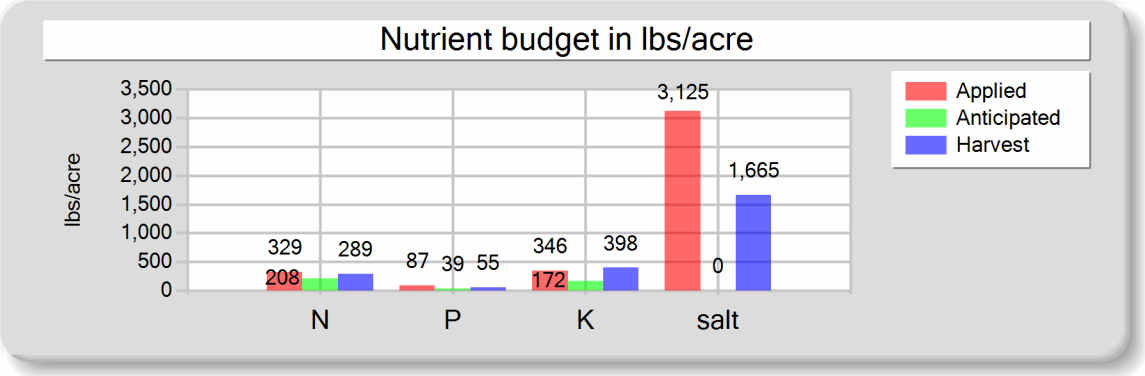
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-12 - 05/28/2023: Corn, silage

Field name: 6-12 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	127.25	70.07	97.37	0.00
Process wastewater	149.36	16.98	248.35	1,435.61
Fresh water	45.66	0.00	0.00	1,689.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	329.28	87.05	345.72	3,125.47
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	289.01	55.19	398.40	1,665.46
Nutrient balance	40.28	31.86	-52.68	1,460.01
Applied to removed ratio	1.14	1.58	0.87	1.88

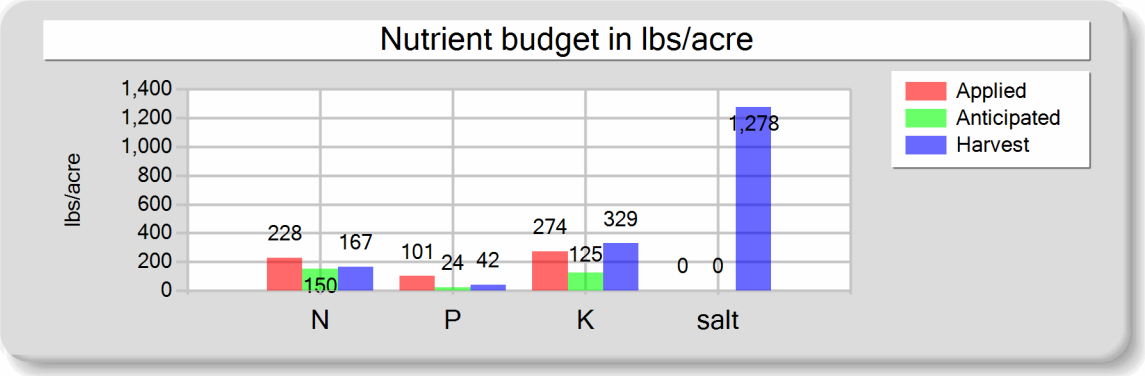
Fresh water applied	
9,000,000.00	*gallons*
331.44	*acre-inches*
33.14	*inches/acre*

Process wastewater applied	
192,000.00	*gallons*
7.07	*acre-inches*
0.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

6-2 - 10/29/2022: Oats, silage-soft dough

Field name: 6-2 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	220.96	101.19	273.96	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.96	101.19	273.96	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	167.46	42.41	328.98	1,278.25
Nutrient balance	60.49	58.77	-55.02	-1,278.25
Applied to removed ratio	1.36	2.39	0.83	0.00

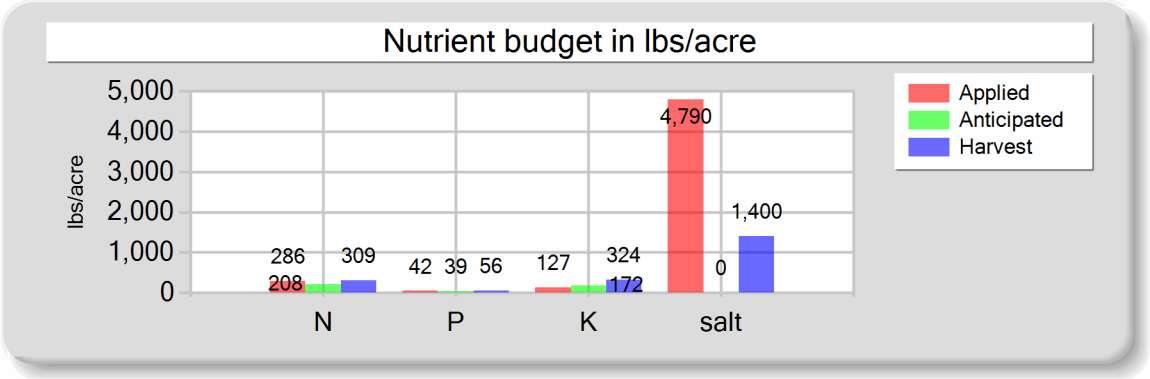
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-2 - 05/28/2023: Corn, silage

Field name: 6-2 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	70.70	38.93	54.10	0.00
Process wastewater	41.26	3.17	72.99	372.97
Fresh water	166.79	0.00	0.00	4,417.01
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.75	42.10	127.09	4,789.97
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	308.68	56.00	323.86	1,400.03
Nutrient balance	-22.93	-13.90	-196.77	3,389.95
Applied to removed ratio	0.93	0.75	0.39	3.42

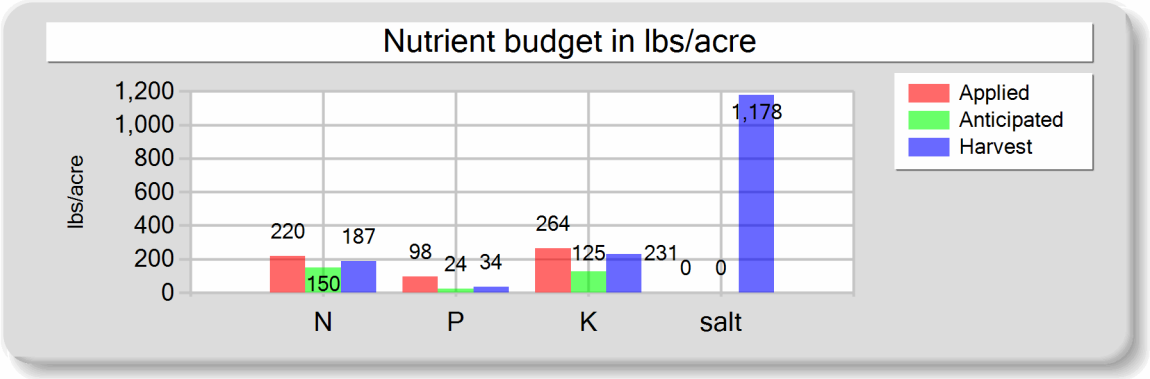
Fresh water applied	
14,220,000.00	*gallons*
523.67	*acre-inches*
29.09	*inches/acre*

Process wastewater applied	
96,000.00	*gallons*
3.54	*acre-inches*
0.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

6-3 - 10/29/2022: Oats, silage-soft dough

Field name: 6-3 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	213.06	97.57	264.17	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.06	97.57	264.17	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	187.20	34.48	230.88	1,177.80
Nutrient balance	32.86	63.10	33.29	-1,177.80
Applied to removed ratio	1.18	2.83	1.14	0.00

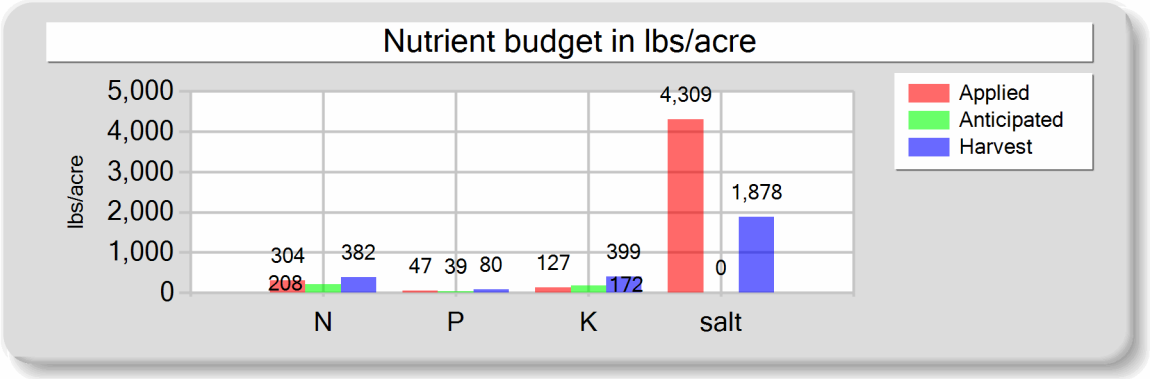
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-3 - 05/28/2023: Corn, silage

Field name: 6-3 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	79.53	43.79	60.86	0.00
Process wastewater	37.13	2.86	65.69	335.67
Fresh water	180.34	0.00	0.00	3,973.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.01	46.65	126.55	4,308.72
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	382.00	79.78	398.88	1,878.35
Nutrient balance	-77.99	-33.13	-272.33	2,430.38
Applied to removed ratio	0.80	0.58	0.32	2.29

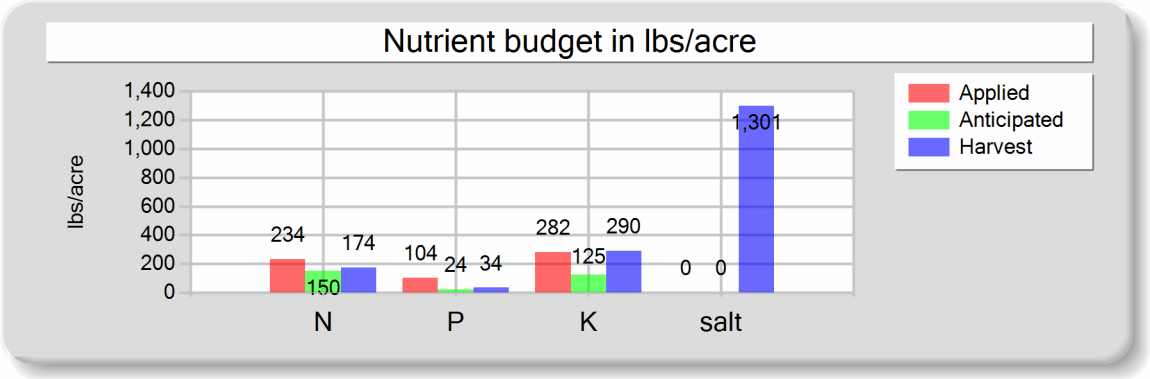
Fresh water applied	
33,120,000.00	*gallons*
1,219.70	*acre-inches*
30.49	*inches/acre*

Process wastewater applied	
192,000.00	*gallons*
7.07	*acre-inches*
0.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

6-4 - 10/29/2022: Oats, silage-soft dough

Field name: 6-4 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	227.27	104.08	281.78	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.27	104.08	281.78	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	173.96	34.23	290.47	1,300.74
Nutrient balance	60.30	69.84	-8.69	-1,300.74
Applied to removed ratio	1.35	3.04	0.97	0.00

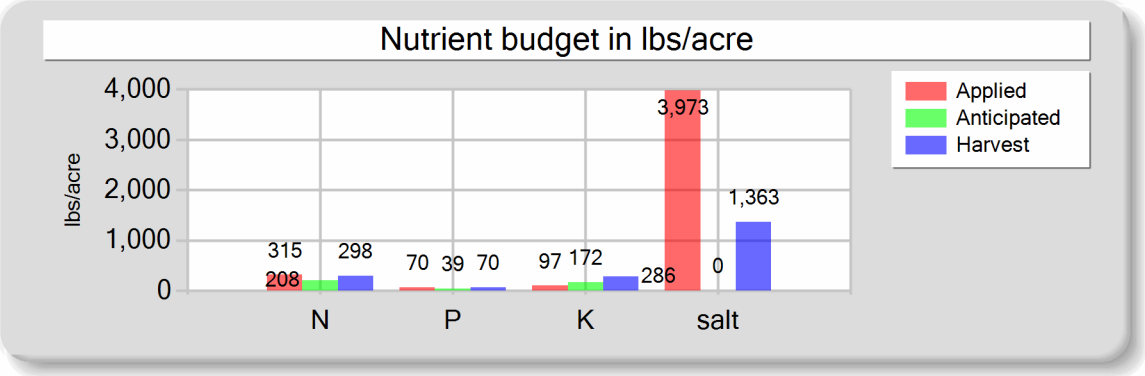
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-4 - 05/28/2023: Corn, silage

Field name: 6-4 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	127.25	70.07	97.37	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	180.34	0.00	0.00	3,973.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.60	70.07	97.37	3,973.05
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	297.88	69.51	286.03	1,363.44
Nutrient balance	16.71	0.56	-188.65	2,609.61
Applied to removed ratio	1.06	1.01	0.34	2.91

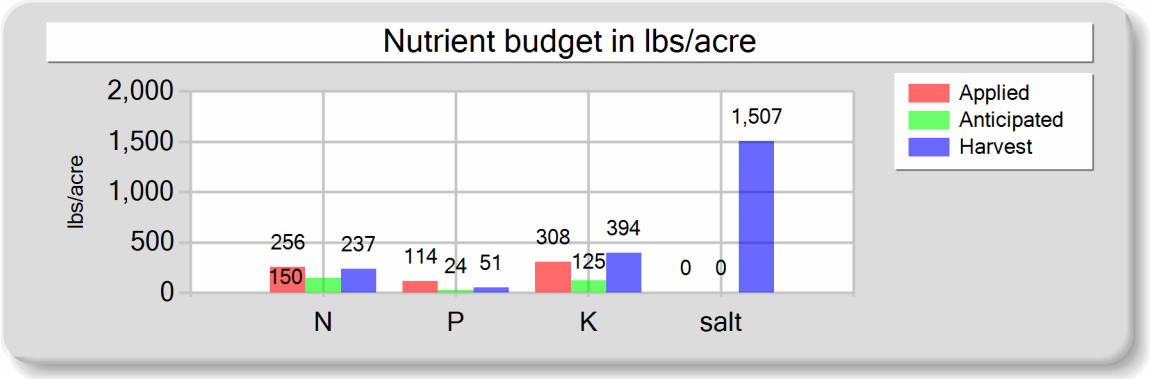
Fresh water applied	
4,140,000.00	*gallons*
152.46	*acre-inches*
30.49	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

6-5 - 10/29/2022: Oats, silage-soft dough

Field name: 6-5 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	248.57	113.83	308.20	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.57	113.83	308.20	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	237.09	50.83	393.81	1,506.94
Nutrient balance	18.48	63.00	-85.61	-1,506.94
Applied to removed ratio	1.08	2.24	0.78	0.00

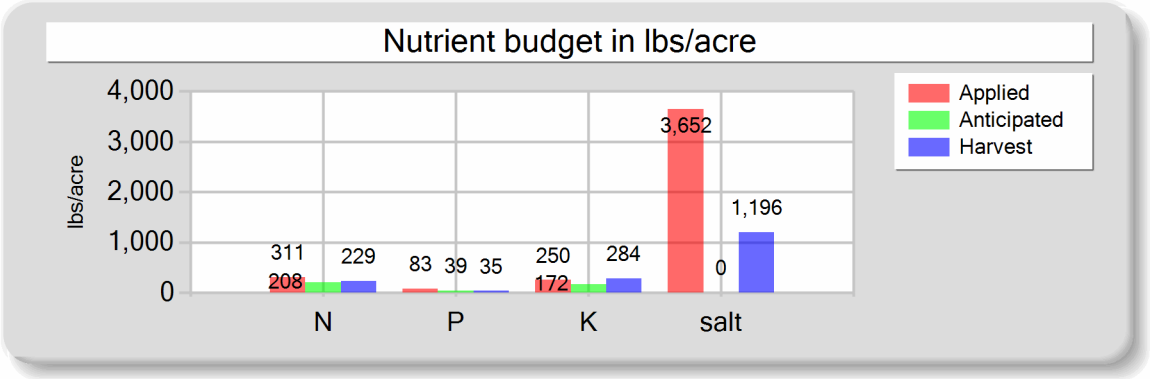
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-5 - 05/28/2023: Corn, silage

Field name: 6-5 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	139.18	76.64	106.50	0.00
Process wastewater	81.23	6.25	143.70	734.28
Fresh water	83.37	0.00	0.00	2,917.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	310.78	82.88	250.20	3,652.11
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	229.00	35.12	284.32	1,196.45
Nutrient balance	81.78	47.76	-34.11	2,455.66
Applied to removed ratio	1.36	2.36	0.88	3.05

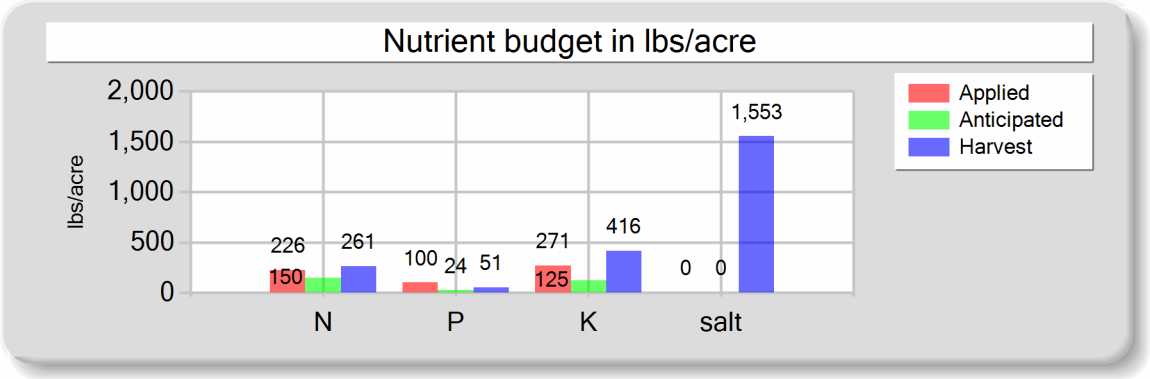
Fresh water applied	
26,640,000.00	*gallons*
981.06	*acre-inches*
30.66	*inches/acre*

Process wastewater applied	
336,000.00	*gallons*
12.37	*acre-inches*
0.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

6-6 - 10/29/2022: Oats, silage-soft dough

Field name: 6-6 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	218.53	100.07	270.95	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.53	100.07	270.95	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	261.29	50.92	415.60	1,553.36
Nutrient balance	-35.77	49.15	-144.66	-1,553.36
Applied to removed ratio	0.86	1.97	0.65	0.00

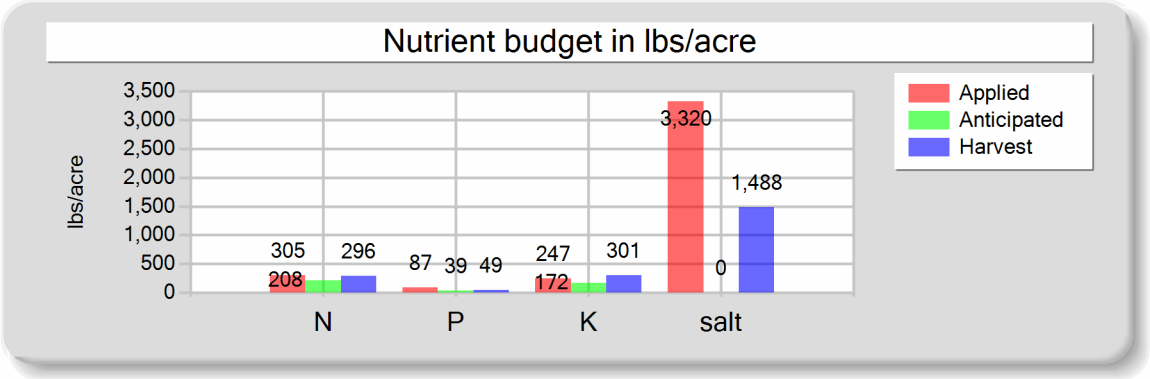
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-6 - 05/28/2023: Corn, silage

Field name: 6-6 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	146.83	80.85	112.35	0.00
Process wastewater	76.17	5.86	134.75	688.55
Fresh water	75.18	0.00	0.00	2,631.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	305.18	86.71	247.11	3,319.92
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	295.87	48.93	300.83	1,487.63
Nutrient balance	9.31	37.78	-53.73	1,832.29
Applied to removed ratio	1.03	1.77	0.82	2.23

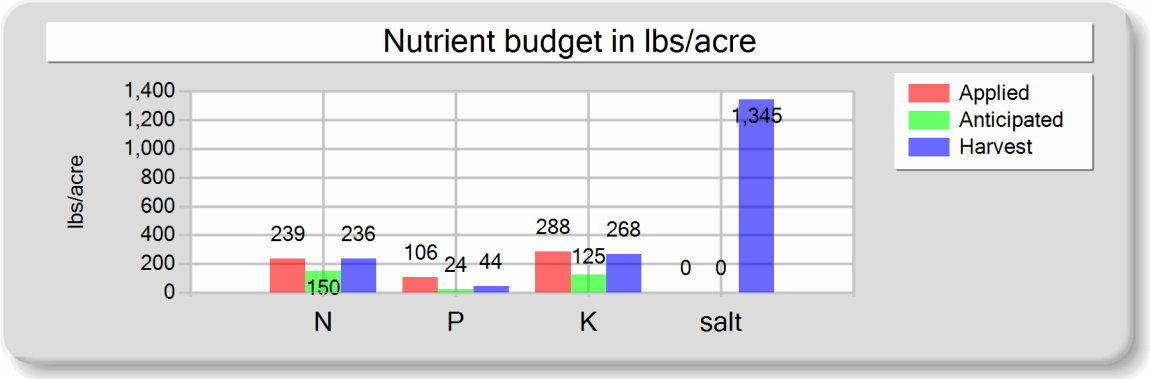
Fresh water applied	
29,280,000.00	*gallons*
1,078.28	*acre-inches*
27.65	*inches/acre*

Process wastewater applied	
384,000.00	*gallons*
14.14	*acre-inches*
0.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

6-7 - 10/29/2022: Oats, silage-soft dough

Field name: 6-7 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	231.91	106.20	287.53	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.91	106.20	287.53	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	236.43	44.12	268.08	1,344.78
Nutrient balance	2.47	62.08	19.46	-1,344.78
Applied to removed ratio	1.01	2.41	1.07	0.00

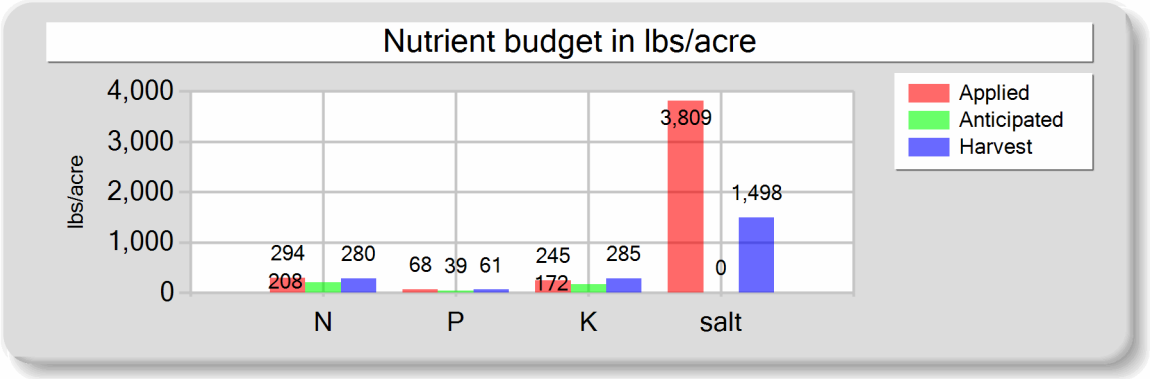
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-7 - 05/28/2023: Corn, silage

Field name: 6-7 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	110.37	60.77	84.46	0.00
Process wastewater	90.94	6.99	160.88	822.05
Fresh water	85.34	0.00	0.00	2,987.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	293.65	67.77	245.33	3,809.08
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	280.45	60.71	285.23	1,497.86
Nutrient balance	13.20	7.06	-39.90	2,311.22
Applied to removed ratio	1.05	1.12	0.86	2.54

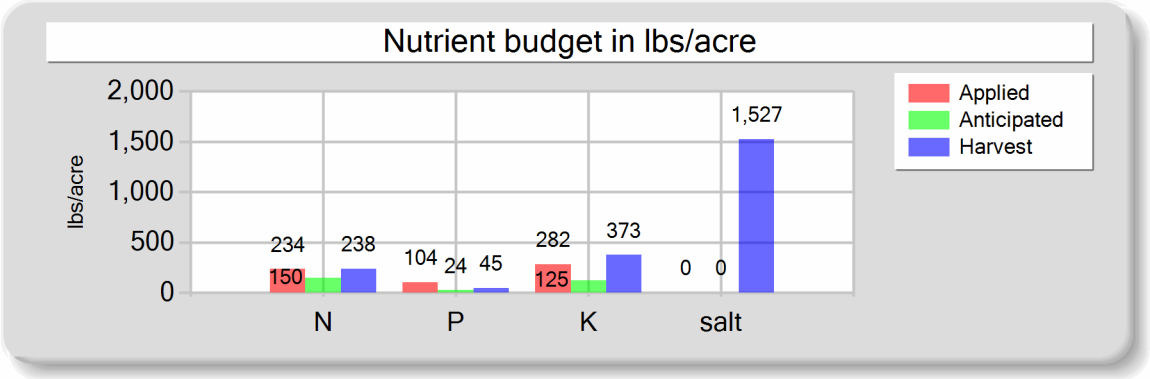
Fresh water applied	
41,760,000.00	*gallons*
1,537.88	*acre-inches*
31.39	*inches/acre*

Process wastewater applied	
576,000.00	*gallons*
21.21	*acre-inches*
0.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

6-8 - 10/29/2022: Oats, silage-soft dough

Field name: 6-8 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	227.27	104.08	281.78	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.27	104.08	281.78	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	238.43	44.65	373.33	1,526.77
Nutrient balance	-4.16	59.42	-91.54	-1,526.77
Applied to removed ratio	0.98	2.33	0.75	0.00

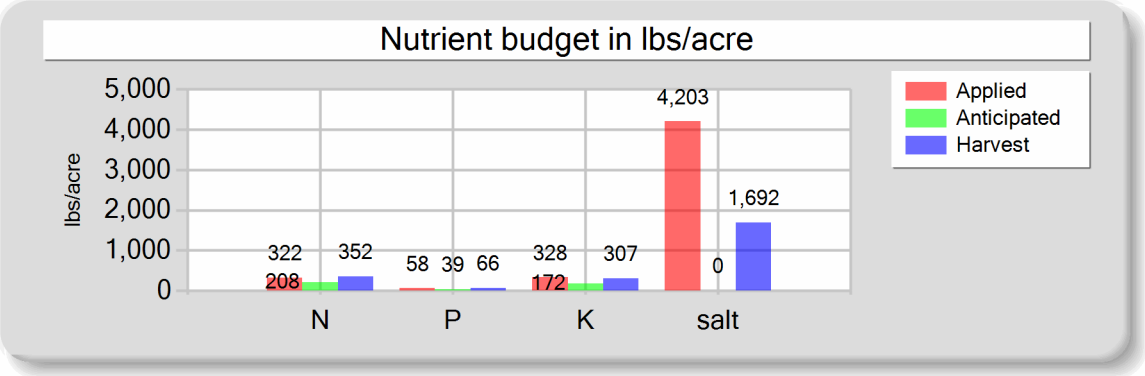
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-8 - 05/28/2023: Corn, silage

Field name: 6-8 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	84.84	46.71	64.92	0.00
Process wastewater	148.53	11.42	262.77	1,342.68
Fresh water	81.71	0.00	0.00	2,860.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	322.08	58.14	327.68	4,202.68
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	352.27	65.73	307.01	1,692.48
Nutrient balance	-30.19	-7.60	20.67	2,510.20
Applied to removed ratio	0.91	0.88	1.07	2.48

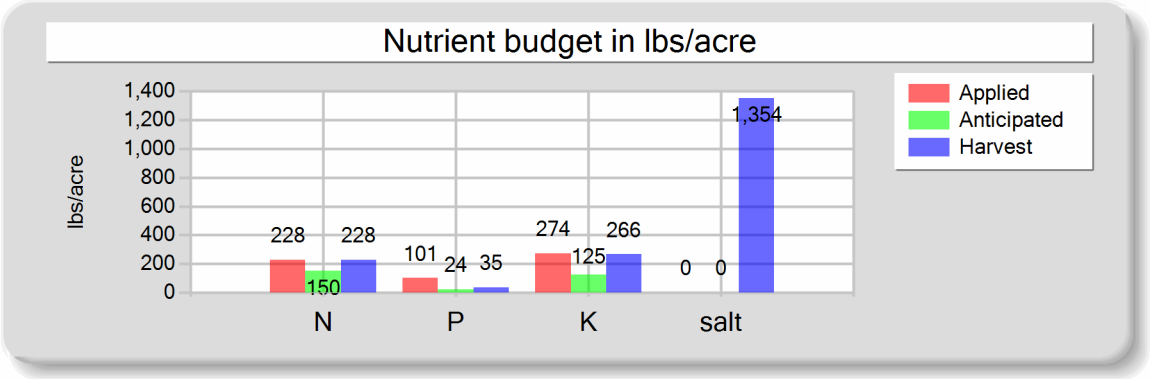
Fresh water applied	
12,240,000.00	*gallons*
450.76	*acre-inches*
30.05	*inches/acre*

Process wastewater applied	
288,000.00	*gallons*
10.61	*acre-inches*
0.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

6-9 - 10/29/2022: Oats, silage-soft dough

Field name: 6-9 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	220.96	101.19	273.96	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.96	101.19	273.96	0.00
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	228.33	34.78	266.39	1,354.05
Nutrient balance	-0.38	66.40	7.57	-1,354.05
Applied to removed ratio	1.00	2.91	1.03	0.00

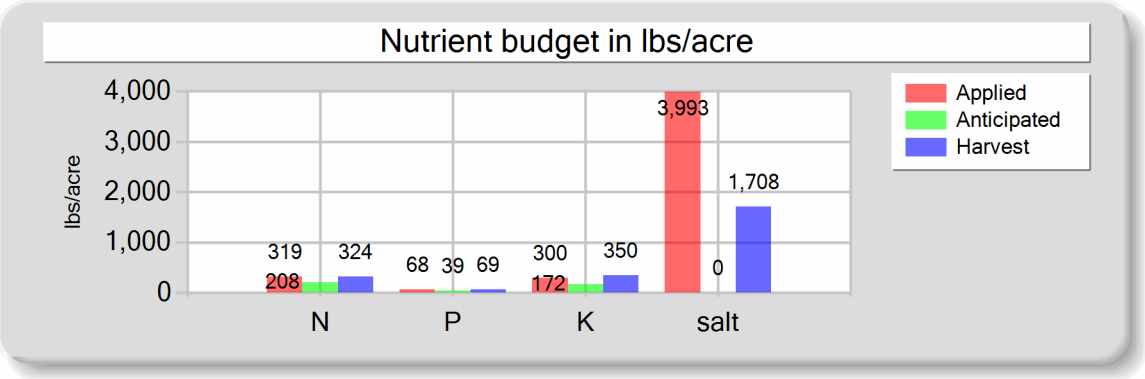
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

6-9 - 05/28/2023: Corn, silage

Field name: 6-9 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	106.05	58.39	81.14	0.00
Process wastewater	123.77	9.52	218.97	1,118.90
Fresh water	82.11	0.00	0.00	2,874.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	318.93	67.91	300.12	3,992.92
Anticipated crop nutrient removal	208.00	39.00	171.60	0.00
Actual crop nutrient removal	323.99	69.12	349.51	1,708.29
Nutrient balance	-5.05	-1.21	-49.40	2,284.62
Applied to removed ratio	0.98	0.98	0.86	2.34

Fresh water applied	
14,760,000.00	*gallons*
543.56	*acre-inches*
30.20	*inches/acre*

Process wastewater applied	
288,000.00	*gallons*
10.61	*acre-inches*
0.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 08/04/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 11.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,100.00	14,700.00	39,800.00							
DL	100.00	100.00	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 01/28/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 35.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,700.00	13,600.00	18,900.00							
DL	100.00	100.00	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 07/27/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 27.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,100.00	6,350.00	39,100.00							
DL	100.00	10.00	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/03/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 12.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,700.00	9,040.00	48,200.00	38,300.00	10,500.00	11,000.00	6,980.00	2,820.00		46.50
DL	100.00	10.00	100.00	100.00	100.00	100.00	10.00	10.00		0.10

B. PROCESS WASTEWATER ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 01/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	638.00	462.00	0.00	0.00	71,800.00	1,250.00								9,630.00	5,280
DL	1.00	1.00	1.00	1.00	1.00	1.00								10.00	10

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 05/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	932.00	616.00	0.00	0.21	106.00	1,550.00								12,000.00	8,960
DL	1.00	1.00	1.00	0.10	1.00	1.00								10.00	10

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	927.00	624.00	0.00	0.00	71.30	1,640.00								12,600.00	8,380
DL	1.00	1.00	1.00	1.00	0.10	10.00								10.00	10

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/03/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	445.00	171.00	0.00	0.00	76.30	978.00	208.00	181.00	373.00	3,320.00	0.00	214.00	361.00	7,310.00	5,360
DL	1.00	1.00	1.00	1.00	0.10	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	10.00	10

C. FRESH WATER ANALYSES

3Dp1

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.08	0.00	6.08								4,060.00	225
DL	0.01	0.10	0.01								10.00	1

3Dp2

3Dp2

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	13.30	0.00	13.30								876.00	415
DL	0.10	1.00	0.10								1.00	1

3Dp3

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	26.10	0.00	26.10								1,130.00	575
DL	0.10	1.00	0.10								10.00	1

3Dp4

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	12.00	0.00	12.00								883.00	420
DL	0.10	0.10	0.10								1.00	1

3Dp5

3Dp5

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	25.30	0.00	25.30								1,340.00	670
DL	0.10	0.10	0.10								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

6-1 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,300.00	3,680.00	31,600.00		13.30
DL	100.00	10.00	100.00		0.10

6-1 - 05/28/2023: Corn, silage

6-1 - 05/28/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,200.00	3,450.00	17,800.00		8.50
DL	100.00	10.00	100.00		0.10

6-11 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,700.00	3,540.00	26,500.00		15.70
DL	100.00	10.00	100.00		0.10

6-11 - 05/28/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 09/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	3,270.00	20,700.00		9.70
DL	100.00	10.00	100.00		0.10

6-12 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,700.00	3,520.00	27,000.00		16.00
DL	100.00	10.00	100.00		0.10

6-12 - 05/28/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,700.00	3,380.00	24,400.00		10.20
DL	100.00	10.00	100.00		0.10

6-2 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,900.00	4,280.00	33,200.00		12.90
DL	100.00	10.00	100.00		0.10

6-2 - 05/28/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	3,320.00	19,200.00		8.30
DL	100.00	10.00	100.00		0.10

6-3 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,000.00	4,420.00	29,600.00		15.10
DL	100.00	10.00	100.00		0.10

6-3 - 05/28/2023: Corn, silage

fFGL Agricultural

Sample and source description: fFGL Agricultural

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,100.00	3,780.00	18,900.00		8.90
DL	100.00	10.00	100.00		0.10

6-4 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,800.00	4,290.00	36,400.00		16.30
DL	100.00	10.00	100.00		0.10

6-4 - 05/28/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,100.00	4,690.00	19,300.00		9.20
DL	100.00	10.00	100.00		0.10

6-5 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,600.00	5,060.00	39,200.00		15.00
DL	100.00	10.00	100.00		0.10

6-5 - 05/28/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/05/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,800.00	2,730.00	22,100.00		9.30
DL	100.00	10.00	100.00		0.10

6-6 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,400.00	4,950.00	40,400.00		15.10
DL	100.00	10.00	100.00		0.10

6-6 - 05/28/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,900.00	2,960.00	18,200.00		9.00
DL	100.00	10.00	100.00		0.10

6-7 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,900.00	5,020.00	30,500.00		15.30
DL	100.00	10.00	100.00		0.10

6-7 - 05/28/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	3,810.00	17,900.00		9.40
DL	100.00	10.00	100.00		0.10

6-8 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,800.00	4,270.00	35,700.00		14.60
DL	100.00	10.00	100.00		0.10

6-8 - 05/28/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,900.00	3,340.00	15,600.00		8.60
DL	100.00	10.00	100.00		0.10

6-9 - 10/29/2022: Oats, silage-soft dough

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,800.00	3,930.00	30,100.00		15.30
DL	100.00	10.00	100.00		0.10

6-9 - 05/28/2023: Corn, silage

FGL Agricultural

Sample and source description: FGL Agricultural

Sample date: 10/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	3,520.00	17,800.00		8.70
DL	100.00	10.00	100.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

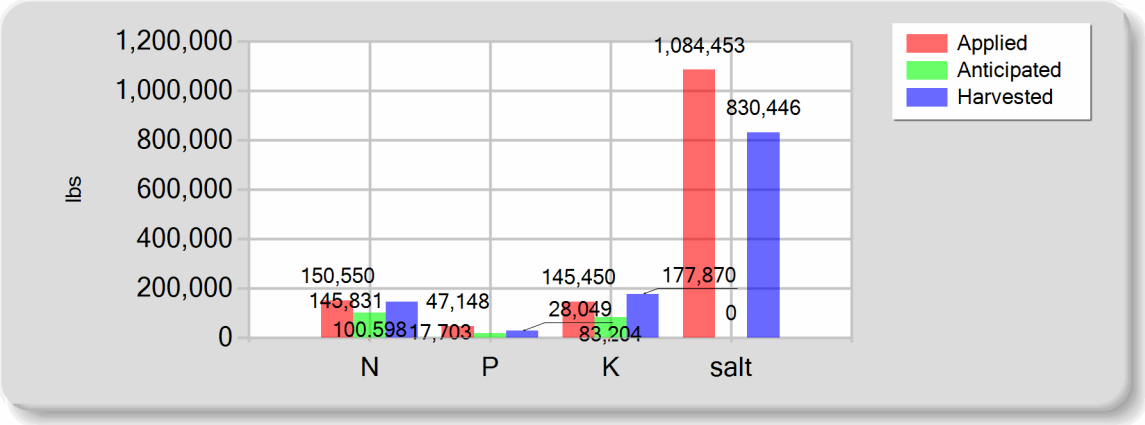
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

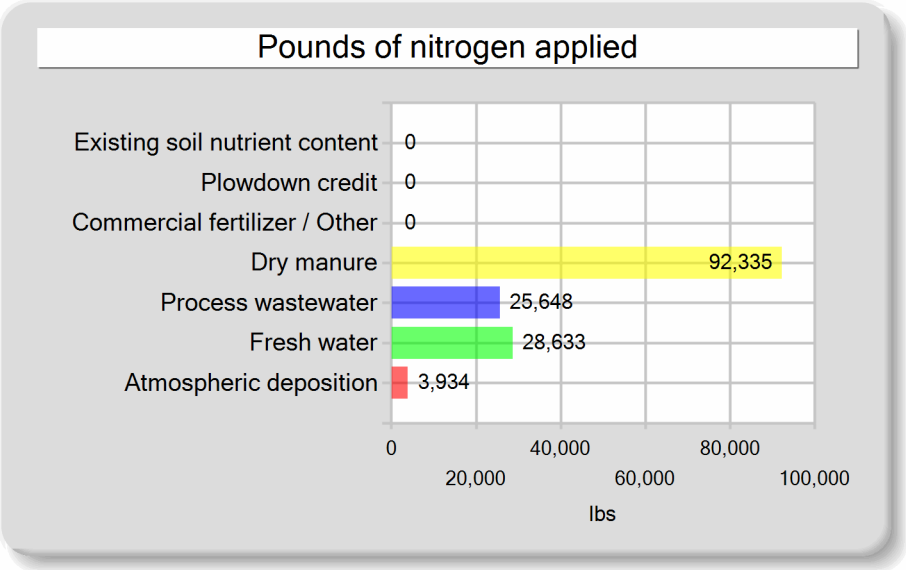
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

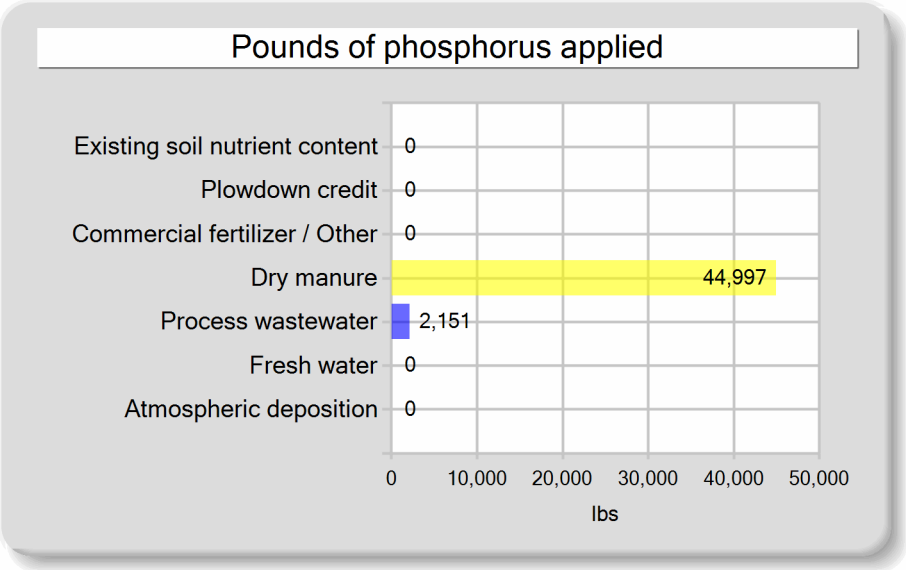
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	92,335.38	44,996.55	100,590.80	0.00
Process wastewater	25,648.15	2,151.33	44,858.71	234,632.03
Fresh water	28,632.82	0.00	0.00	849,820.59
Atmospheric deposition	3,934.00	0.00	0.00	0.00
Total nutrients applied	150,550.35	47,147.87	145,449.52	1,084,452.61
Anticipated crop nutrient removal	100,598.00	17,703.00	83,204.10	0.00
Actual crop nutrient removal	145,830.65	28,048.74	177,869.57	830,446.44
Nutrient balance	4,719.70	19,099.13	-32,420.05	254,006.17
Applied to removed ratio	1.03	1.68	0.82	1.31

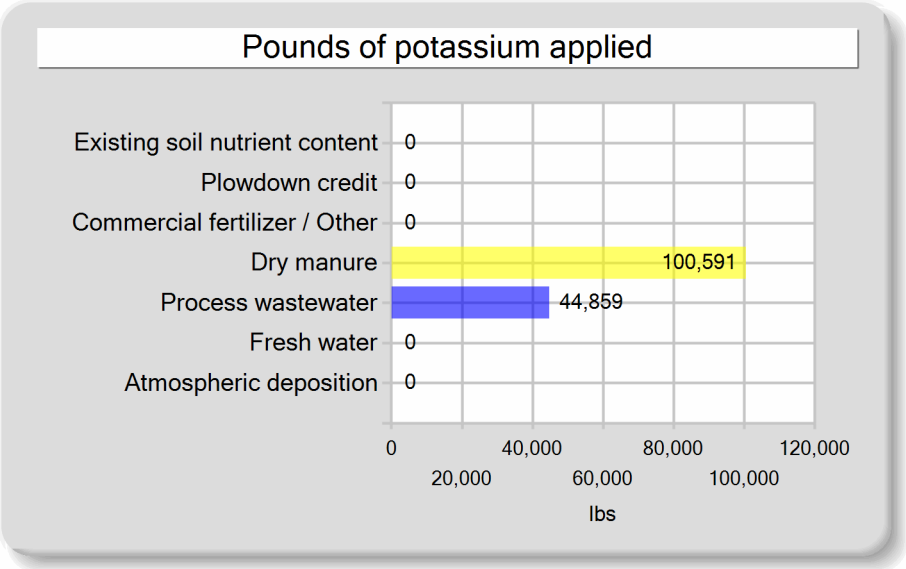
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

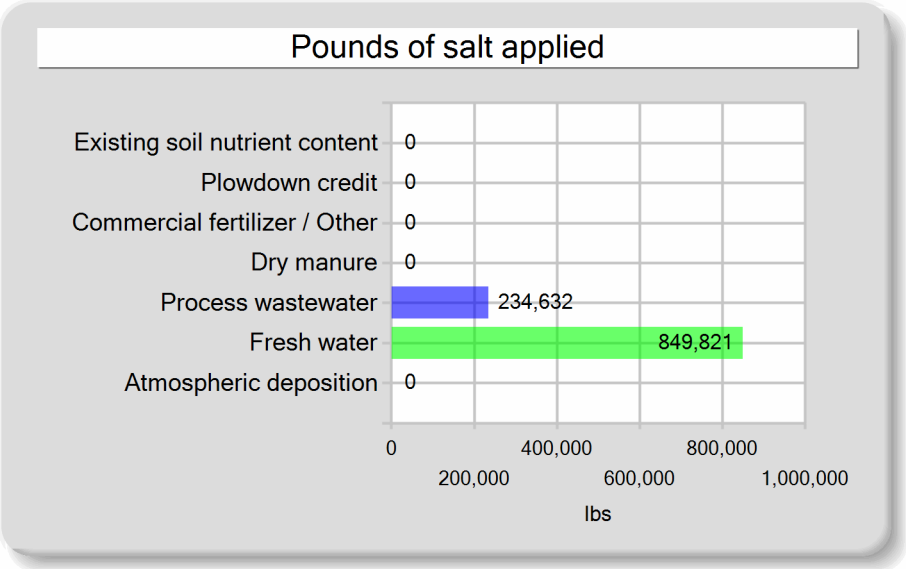


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Machado Dairy Farms #3	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

X Amaral L. Machado
SIGNATURE OF OWNER OF FACILITY

Machado Dairy Farms #3
PRINT OR TYPE NAME

6-4-24
DATE

Amaral L. Machado
SIGNATURE OF OPERATOR OF FACILITY

SAME AS OWNER
PRINT OR TYPE NAME

6-4-24
DATE

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Machado Dairy Farms #3

Name of Dairy Facility: Machado Dairy Farms #3

Facility Address:

Number and Street	City	County	Zip Code
11685 E Mariposa RD	Stockton	San Joaquin	95215

Contact Person Name and Phone Number: Donald Machado (Name) (209) 602-3474 (Phone Number)

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Machado Dairy Farms

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 4430	Manteca	CA	95337

Contact Person: John Machado (Name) (209) 602-8598 (Phone Number)

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Mossdale Associates (Name) (209) 602-8598 (Phone Number)

Address	City	State	Zip Code
800 W Mossdale RD	Manteca	CA	95337

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
800 W Mossdale RD	Manteca	95337

Street and nearest cross street (if no address) — County: San Joaquin

Assessor's Parcel Number: 1111-1111-1111-1111 — Assessor's Parcel Number County: San Joaquin

Last date hauled: 11/08/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 3,760.00 tons

Manure Solids Content: 87.8 %

Method used to determine amount of manure:

Weight per load times number of loads.

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

X Israel A. Machado — 6-4-24
Operator Signature — Date

John A Machado — 6-4-24
Hauler Signature — Date

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Machado Dairy Farms #1,2,3 & 4
P.O. Box 4430
Manteca, Ca 95337

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER: #3
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 28.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
3D1	T3208008C	Electrical Conductivity (EC)	EPA 120.1 and	0.406	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	6.08	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	225	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	6.08	mg/L
		Soluble Salts (SALT-SOL)	Calculation	260	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	102	104 (T3208033A)	104 (T3208033A)			0.538
Total Dissolved Solids (TDS)	102			225 (T3208008C)	225 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which th customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Machado Dairy Farms #1,2,3 & 4
P.O. Box 4430
Manteca, Ca 95337

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER: #3
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 28.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
3D2	T3208008D	Electrical Conductivity (EC)	EPA 120.1 and	1.58	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	38.4	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	800	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	38.4	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1010	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	102	104 (T3208033A)	104 (T3208033A)			0.538
Total Dissolved Solids (TDS)	98.0			50.0 (T3208034A)	50.0 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which th customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Machado Dairy Farms #1,2,3 & 4
P.O. Box 4430
Manteca, Ca 95337

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER: #3
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 28.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
3Dp1	T3208008E	Electrical Conductivity (EC)	EPA 120.1 and	1.13	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.1	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	555	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	26.1	mg/L
		Soluble Salts (SALT-SOL)	Calculation	724	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	102	104 (T3208033A)	104 (T3208033A)			0.538
Total Dissolved Solids (TDS)	98.0			50.0 (T3208034A)	50.0 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Machado Dairy Farms #1,2,3 & 4
P.O. Box 4430
Manteca, Ca 95337

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER: #3
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 28.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
3Dp2	T3208008F	Electrical Conductivity (EC)	EPA 120.1 and	0.876	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	13.3	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	415	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	13.3	mg/L
		Soluble Salts (SALT-SOL)	Calculation	561	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	102	104 (T3208033A)	104 (T3208033A)			0.538
Total Dissolved Solids (TDS)	98.0			50.0 (T3208034A)	50.0 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

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Compliance Analysis Report

Customer
Machado Dairy Farms #1,2,3 & 4
P.O. Box 4430
Manteca, Ca 95337

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER: #3
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 26.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
3Dp3	T3208008G	Electrical Conductivity (EC)	EPA 120.1 and	1.13	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.1	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	575	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	26.1	mg/L
		Soluble Salts (SALT-SOL)	Calculation	723	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	102	104 (T3208033A)	104 (T3208033A)			0.538
Total Dissolved Solids (TDS)	98.0			50.0 (T3208034A)	50.0 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

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Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER: #3
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 26.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
3Dp4	T3208008H	Electrical Conductivity (EC)	EPA 120.1 and	0.883	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	12.0	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	420	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	12.0	mg/L
		Soluble Salts (SALT-SOL)	Calculation	565	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	102	104 (T3208033A)	104 (T3208033A)			0.538
Total Dissolved Solids (TDS)	98.0			50.0 (T3208034A)	50.0 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

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Galt, CA 95632

PURCHASE ORDER: #3
RECEIVED DATE: 7/27/23 12:35 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 7/28/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 26.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
3Dp5	T3208008I	Electrical Conductivity (EC)	EPA 120.1 and	1.34	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	25.3	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	670	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	25.3	mg/L
		Soluble Salts (SALT-SOL)	Calculation	854	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.8			0.000 (T3208006A)	0.000 (null)	
Electrical Conductivity (EC)	98.6			6.76 (T3208009B)	6.76 (Dup)	0.089
Nitrate Nitrogen (NO3-N)	102	104 (T3208033A)	104 (T3208033A)			0.538
Total Dissolved Solids (TDS)	98.0			50.0 (T3208034A)	50.0 (Dup)	0.000
Total Nitrogen (TN)	95.2			2.80 (T3208007E)	2.80 (null)	0.000

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.
865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: MACHADO DAIRY FARMS #3
Mailing address: P.O. BOX 4430
(not facility address) MANTECA, CA 95337
Phone#: 209-825-1913

Authorized Copy Released To: BRUCE GOLDSMITH

Sampled By: BRUCE GOLDSMITH
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	7-25-23	1:00	MANURE COMPOST						X		
2		1:10	WASTE WATER			28.6					H-4
3		1:20	3D1	DOMESTIC WELL		28.0					H-7
4		1:25	3D2	—"—		28.0					H-7
5		1:35	3Dp1	IRRIGATION WELL		28.0					H-7
6		1:45	3Dp2	—"—		28.0					H-7
7		2:00	3Dp3	—"—		26.6					H-7
8		2:10	3Dp4	—"—		26.6					H-7
9		2:20	3Dp5	—"—		26.6					H-7
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[illegible]	DAIRY COMPLIANCE GROUP				
2nd		DENELE ANALYTICAL, INC.	7-27-23 12:35			T3208008

Required ~~to be submitted~~ to any agency? (YES / NO)